

BAKTI DWI PAMUNGKAS

Informatics Engineering Student · AI & Machine Learning Enthusiast

Sumedang, Indonesia | 089656428641 | baktidwipamungkas@gmail.com | linkedin.com/in/baktidwipamungkas | bakti-ai.vercel.app | github.com/BAKTIDWIPAMUNGKAS

EDUCATION

Informatics Engineering (S1) | Universitas Siliwangi

2023 – Present

- ▶ Active member and contributor in HMIF FT-UNSIL student organization
- ▶ Focused coursework in algorithms, data structures, and software engineering

EXPERIENCE

Machine Learning Engineer Intern | PT Sawala Technology

June 2026 – August 2026

- ▶ Engineered an automated, end-to-end Hybrid Data Pipeline (V2.2) for the Kelas Tempo educational platform, processing over 1.1 million transaction log records
- ▶ Resolved critical Out of Memory (OOM) exceptions and network bottlenecks by migrating the ETL infrastructure to a native WSL2 Ubuntu environment
- ▶ Orchestrated data workflows using Apache Airflow (DAGs) and executed large-scale data cleansing with PySpark, refining 15,000 raw rows into 4,771 financially valid records using strict Inner Join operations
- ▶ Developed and deployed a Hybrid Rule-Based Inference and K-Nearest Neighbors (KNN) classification model for real-time customer segmentation, achieving 99.80% global accuracy on the production backend
- ▶ Designed an interactive Customer Profiling Dashboard in Tableau to translate RFM metrics and course preferences into actionable cross-selling strategies for the management team

PROJECTS

Plantify – AI-Powered Plant Disease Diagnosis App | Personal Project

2026

- ▶ Built a full-stack web application enabling farmers to diagnose crop pests and diseases by uploading plant photos
- ▶ Integrated multimodal LLM via OpenRouter API with structured prompt engineering to return expert-level diagnosis including pest/disease name, severity level, recommended pesticide, dosage, and treatment steps
- ▶ Designed a comprehensive JSON-based prompt schema to ensure consistent and actionable AI outputs for agricultural use cases
- ▶ Tech stack: Python, Flask, OpenRouter API (Gemma Vision), HTML/CSS/JavaScript

Sentiment Analysis: LSTM vs Naive Bayes on Iran-Israel Conflict Tweets | Personal Project

2026

- ▶ Scraped and curated a dataset of 2,000+ tweets on the Iran-Israel conflict using Playwright-based Twitter crawler with multi-keyword and time-window strategies
- ▶ Built full NLP preprocessing pipeline including translation, tokenization, stopword removal, and stemming using Sastrawi
- ▶ Trained and compared LSTM with word embeddings vs TF-IDF + Naive Bayes across accuracy, F1-score, and confusion matrix for 3-class sentiment classification
- ▶ Tech stack: Python, TensorFlow/Keras, Scikit-learn, Pandas, NLTK, Sastrawi, Matplotlib, Google Colab

SKILLS

Languages: Python

Frameworks & Libraries: PyTorch, TensorFlow, Keras, Scikit-learn, Flask, PySpark, NumPy, Pandas, OpenCV, Matplotlib, Seaborn

Concepts: Machine Learning, Deep Learning, NLP, Computer Vision, Sentiment Analysis

Tools: Git, GitHub, Google Colab, Visual Studio Code, Jupyter Notebook, Apache Airflow, Tableau

Soft Skills: Team collaboration, critical thinking

CERTIFICATIONS & TRAINING

- ▶ Machine Learning Specialization – Coursera / University of Washington (2025)